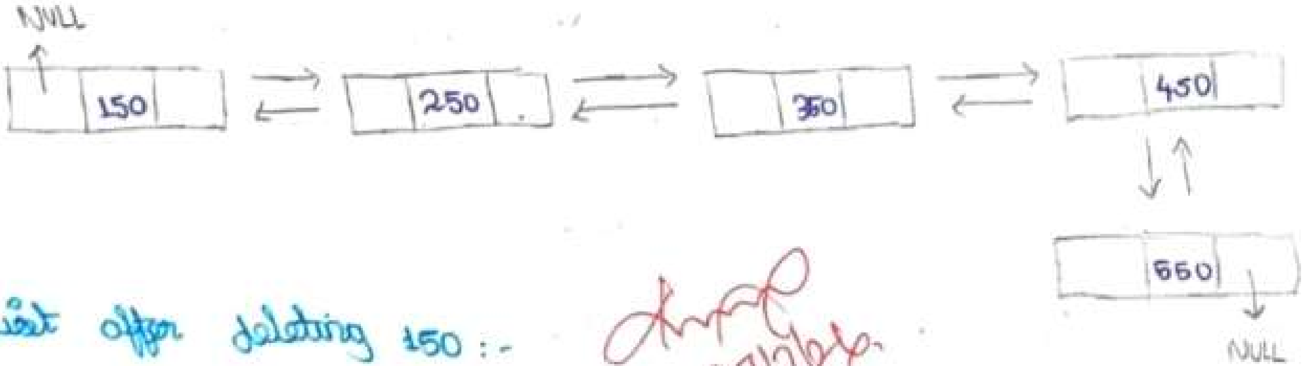
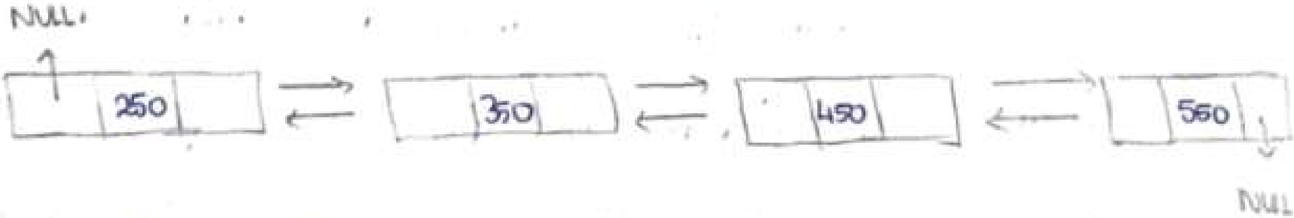
Co-2 Assessment - 1

Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the element 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after operation:-

Doubly Linked List after insertion:-

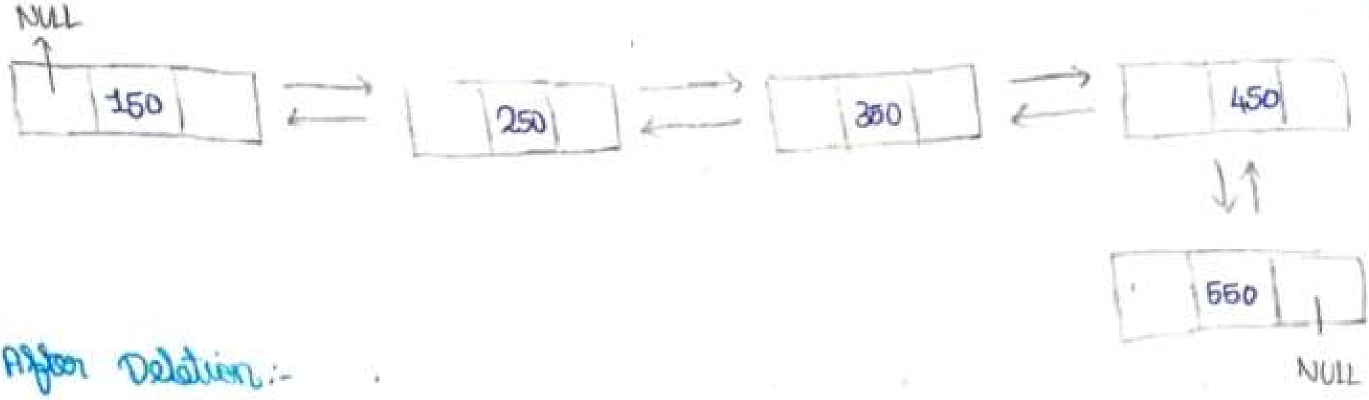


List after deleting 150:-



List after deleting 350:-

Before Deletion :-

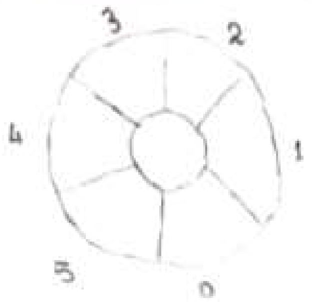


After Deletion :-

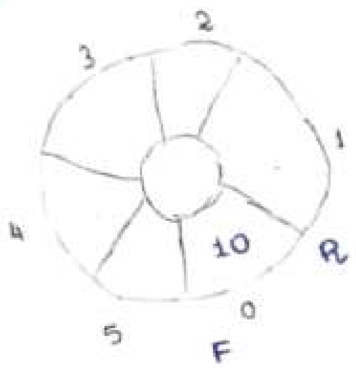
2. A Circular queue has a size of 6. Execute the following operations : Enqueue (10), Enqueue (20), Enqueue (30), Dequeue (), Enqueue (40), Enqueue (50), Enqueue (60), Enqueue (70), Draw the queue after each operation and identify the values of the front and rear pointers

A Circular queue is a queue in which the last position is connect to the first position. when the rear reaches the last index, it wraps around to the beginning if there is free space. It follows the FIFO (First in First out) principle,

Queue size = 6 (indices : 0 to 5)



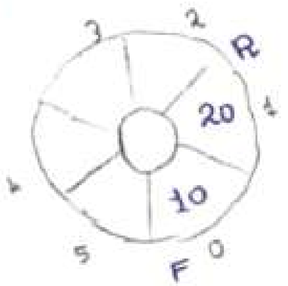
Enqueue (10):



Front = 0

Rear = 0

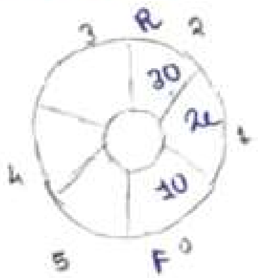
Enqueue (20)



Front = 0

Rear = 1

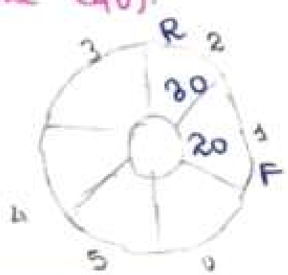
Enqueue (30)



Front = 0

Rear = 2

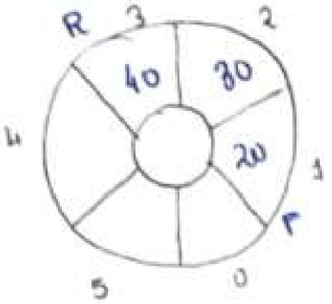
Dequeue (40):



Front = 1

Rear = 2

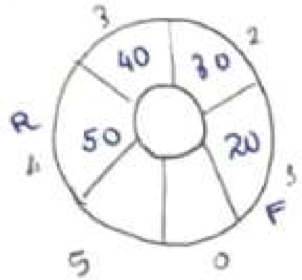
Enqueue (40)



Front = 1

Rear = 3

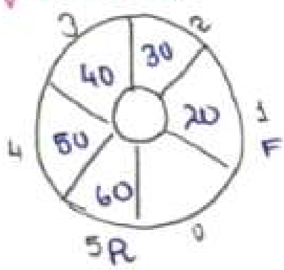
Enqueue (50)



Front = 1

Rear = 4

Enqueue (60)

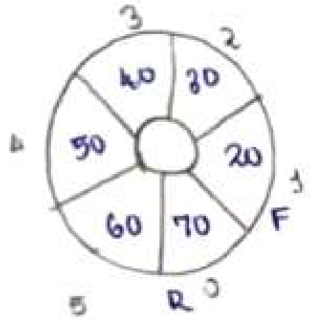


Front = 1

Rear = 4

Enqueue (70)

Since the rear reaches the last index (5), it wraps around to index 0 and insert 70.



Front = 1

Rear = 0